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Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

OPEN ENDED Report

on

THROTTLE - BY - WIRE

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Under the Guidance of

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled **"Throttle By Wire"** is a bonafide work carried out by the student team of **"Chandrashekhar Bhavi-02FE22BEC020, Jamuna Pulpatli-02FE22BEC032, Priyanka Budihal-02FE22BEC054,Pruthviraj Banoshi-02FE22BEC058"**.The project report has been approved as it satisfies the requirements with respect to the minor project work prescribed by the university curriculum for B.E. (VI Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2025-2026.

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ABSTRACT

This project demonstrates the design and implementation of a basic Throttle-by-Wire (TBW) system using an ESP32 microcontroller. The traditional mechanical linkage between the accelerator pedal and throttle valve is replaced with an electronic system consisting of a potentiometer to simulate the accelerator input and a servo motor to control throttle position. The ESP32 reads the analog input from the potentiometer, maps it to a corresponding angle, and commands the servo motor to adjust accordingly. This setup mimics the behavior of modern electronic throttle control systems found in vehicles. The project highlights key concepts of drive-by-wire technology, such as sensor integration, PWM control, signal processing, and real-time actuation. The implementation showcases the potential of embedded systems in developing automotive electronics and serves as a foundational step towards advanced vehicle automation.

Contents

Chapter 1

Introduction

1.1 Problem Statement

Design a Throttle by Wire system using the ESP32 microcontroller to electronically control engine throttle based on accelerator pedal input. The system must ensure accurate sensor reading, fast and reliable motor actuation, and real-time response. Key challenges include ensuring safety, fault tolerance, and stable control under varying conditions. The goal is to build a cost-effective and robust solution for embedded automotive control.

1.2 Introduction

In modern automotive systems, electronic control is increasingly replacing traditional mechanical components to enhance performance, safety, and fuel efficiency. One such innovation is the Throttle-by-Wire (TBW) system, also known as Electronic Throttle Control (ETC). Unlike conventional throttle systems that rely on a mechanical linkage between the accelerator pedal and the throttle valve, TBW systems use electronic sensors, control units, and actuators to manage engine throttle position.

This experiment focuses on designing and simulating a basic Throttle-by-Wire system. The accelerator pedal input is detected using a sensor and processed by a microcontroller or ECU, which in turn drives a motor to adjust the throttle valve. Feedback from the throttle position sensor is used to ensure accurate valve control through a closed-loop system.

The experiment aims to demonstrate how TBW systems enhance engine response, allow integration with advanced driver-assistance systems, and reduce mechanical complexity. Understanding this system is essential for students and engineers working on modern automotive electronics and autonomous vehicle technology.

1.3 Objectives

1.To understand the working principle of a Throttle-by-Wire (TBW) system and how it replaces mechanical throttle control with electronic components.

2.To design and simulate a basic TBW system using sensors, actuators, and a control algorithm.

3.To implement closed-loop control for maintaining accurate throttle valve positioning based on accelerator pedal input.

4.To analyze the system response to different input conditions and evaluate performance in terms of accuracy and responsiveness.

5.To explore the advantages and practical applications of TBW systems in modern vehicles, including integration with cruise control and autonomous driving systems.

1.4 Applications

- **Adaptive Cruise Control (ACC):** Allows automatic adjustment of vehicle speed and throttle based on traffic conditions.
- **Traction and Stability Control Systems:** Works with other systems to manage engine power and prevent wheel slip.
- **Autonomous and Semi-Autonomous Driving:** Essential for automated throttle control in self-driving vehicles.
- **Fuel Efficiency Optimization:** Enables precise throttle management for improved fuel economy and reduced emissions.
- **Drive Mode Selection:** Allows dynamic changes in throttle response based on selected drive modes (e.g., Eco, Sport).
- **Start/Stop Engine Systems:** Supports automatic engine shutoff and restart to save fuel when the vehicle is idle.

Chapter 2

Design Specification

2.1 Simulation

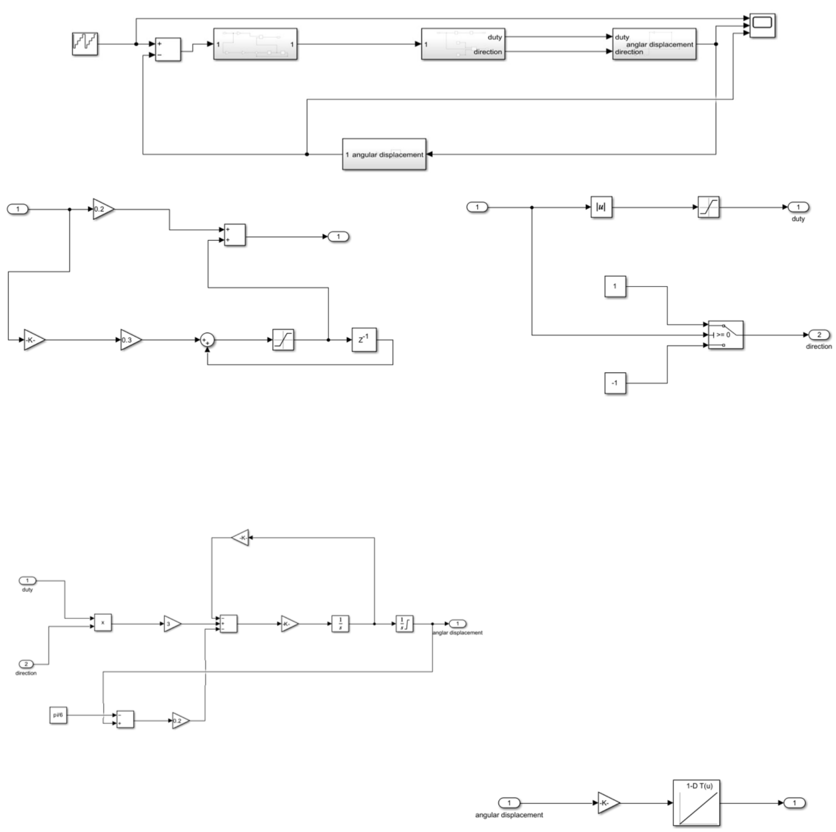

Figure 2.1: Simulation

This Simulink model represents a basic Throttle-by-Wire (TBW)control system, commonly used in automotive applications to electronically control the throttle valve. The system takes a reference input (simulating accelerator pedal position), calculates the error with respect to the current angular position, and processes it through a controller (likely PI or PID). The output is converted into a PWM duty cycle and a direction signal, which control a simulated motor or actuator. Feedback from the motor's angular displacement is used to achieve closed-loop control. Additional logic in the model ensures proper handling of signal magnitude and direction, allowing the system to adjust throttle position accurately and smoothly.

2.2 Hardware

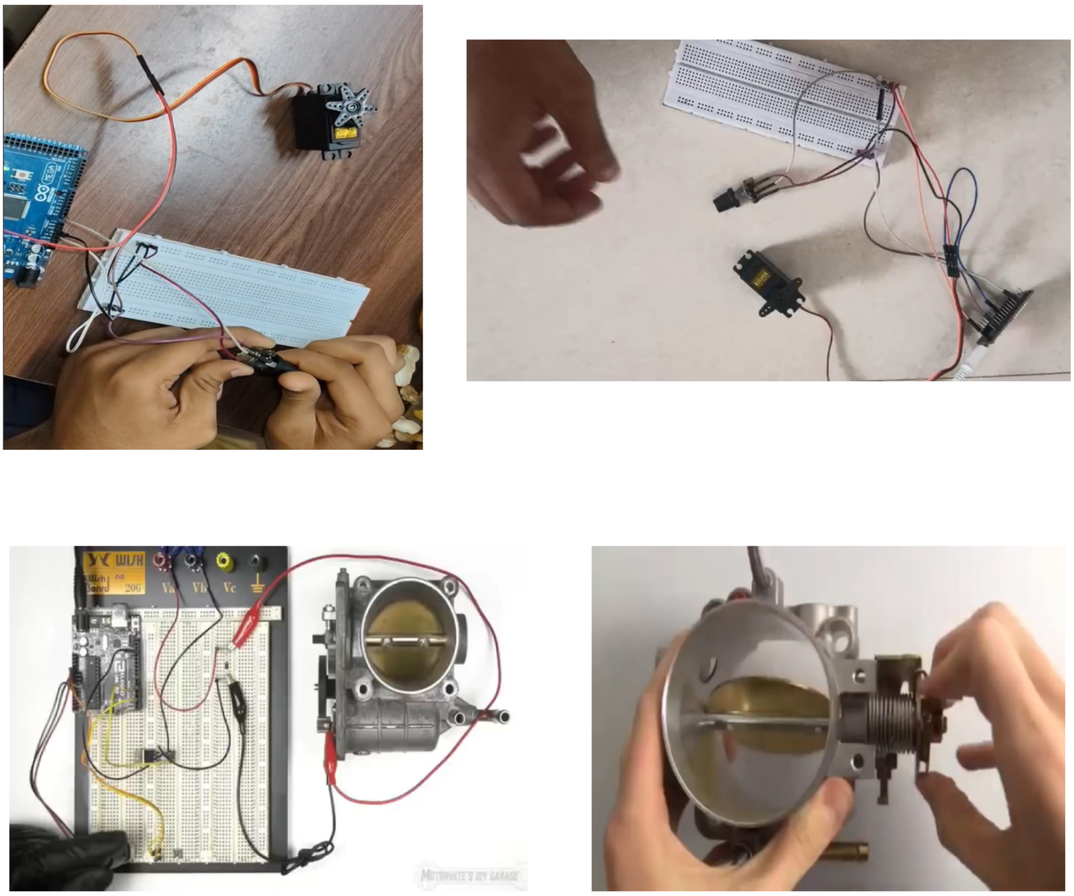

Figure 2.2: Hardware model of Throttle-by-Wire

2.3 Code

```
include ESP32Servo.h / Use ESP32-compatible Servo library
const int potPin = 34; // Potentiometer connected to GPIO34 (ADC1-5)
const int servoPin = 18; // Servo signal pin connected to GPIO18 (PWM-capable)
Servo myServo; void setup()
Serial.begin(115200); // Start Serial Monitor
myServo.setPeriodHertz(50);// Set servo PWM frequency to 50 Hz
myServo.attach(servoPin, 500, 2400);// Attach servo (min/max pulse in µs)
void loop()
int potValue = analogRead(potPin); // Read analog value (0–4095)
int angle = map(potValue, 0, 4095, 0, 180); // Map to 0–180° angle
myServo.write(angle);// Move servo to angle
Serial.print(”Potentiometer Value: ”);
Serial.print(potValue);
Serial.print(” — Servo Angle: ”);
Serial.println(angle);
delay(100); // Delay for stability
```

Chapter 3

Result and Conclusion

3.1 Result

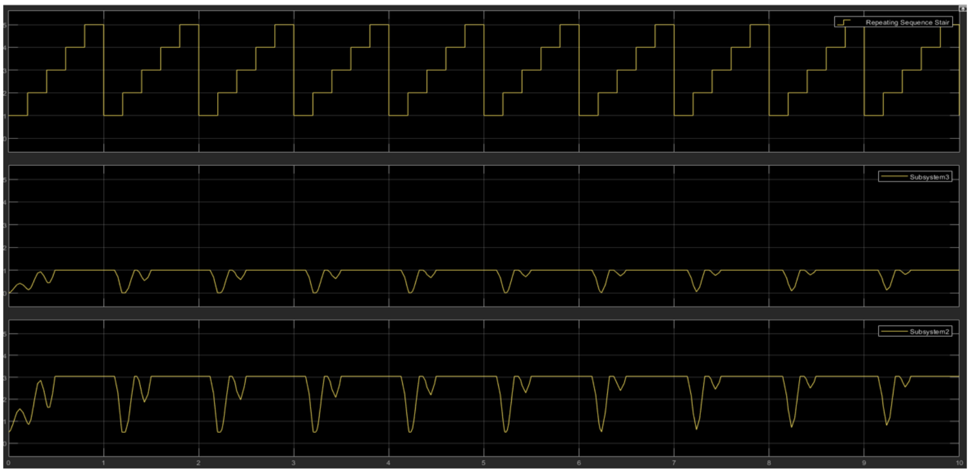

Figure 3.1: Output

The simulation output displays the response of a Throttle-by-Wire system using a closed-loop control strategy. The top plot shows a repeating staircase waveform that represents the desired throttle input, simulating step-wise increases and resets like a driver's accelerator pedal movement. The middle plot represents the system's control or actuator response, which attempts to follow the input signal but exhibits slight oscillations—indicative of real-time system dynamics and minor overshoots. The bottom plot shows the motor's angular displacement or the control error, which also mirrors the input but with more pronounced dips, reflecting rapid corrective actions from the controller. Overall, the plots indicate that the system effectively tracks the input with some transient behavior, highlighting the importance of proper tuning in the controller to ensure smooth and stable throttle operation.

3.2 Conclusion

The implementation of a Throttle by Wire system using the ESP32 microcontroller demonstrates a reliable and efficient approach to modern automotive throttle control. By replacing mechanical linkages with electronic signals, the system improves response time, integration capabilities, and overall vehicle performance. The ESP32 provides a cost-effective platform for real-time control, sensor integration, and actuator management. This project successfully highlights the potential of embedded systems in enhancing automotive safety, efficiency, and functionality.